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Interactions and Correlations in Condensed Matter

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Introduction

1 Second Quantization

In the first quantization formalism, we describe a system of N particles by a wavefunction $\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)$, which is a function of the coordinates of all the particles. This wavefunction must satisfy the appropriate symmetry properties, depending on whether we are dealing with bosons or fermions. For bosons, the wavefunction must be symmetric under the exchange of any two particles, while for fermions, the wavefunction must be antisymmetric under the exchange of any two particles:

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_i, \ldots, \mathbf{r}_j, \ldots, \mathbf{r}_N) = \begin{cases} +\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_j, \ldots, \mathbf{r}_i, \ldots, \mathbf{r}_N) & \text{for bosons,} \\ -\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_j, \ldots, \mathbf{r}_i, \ldots, \mathbf{r}_N) & \text{for fermions.} \end{cases}$$

But an important limitation of this formalism is that it cannot describe systems with a variable number of particles, since <u>the wavefunction is defined for a fixed number of particles</u>

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N) = \sum_{\nu_1 \ldots \nu_N} B_{\nu_1 \ldots \nu_N} \, S_{\pm} \phi_{\nu_1}(\mathbf{r}_1) \phi_{\nu_2}(\mathbf{r}_2) \cdots \phi_{\nu_N}(\mathbf{r}_N),$$

where $S_{\pm}$ is the symmetrization or antisymmetrization operator, depending on whether we are dealing with bosons or fermions, and $\phi_{\nu}(\mathbf{r})$ are the single-particle wavefunctions. The coefficients $B_{\nu_1 \ldots \nu_N}$ are the expansion coefficients of the wavefunction in terms of the single-particle states. Thus, the wavefunction is defined for a fixed number of particles N, and it cannot describe processes that involve the creation or annihilation of particles. Furthermore we can write

$$S_{\pm} \phi_{\nu_1}(\mathbf{r}_1) \phi_{\nu_2}(\mathbf{r}_2) \cdots \phi_{\nu_N}(\mathbf{r}_N) = \begin{pmatrix} \psi_{\nu_1}(\mathbf{r}_1) & \cdots & \psi_{\nu_1}(\mathbf{r}_N) \\ \vdots & \ddots & \vdots \\ \psi_{\nu_N}(\mathbf{r}_1) & \cdots & \psi_{\nu_N}(\mathbf{r}_N) \end{pmatrix}_{\pm},$$

since the symmetrization or antisymmetrization operator can be expressed as a sum over all permutations of the particles, with a plus sign for bosons and a minus sign for fermions, which is equivalent to the determinant or permanent of the matrix of single-particle wavefunctions, depending on whether we are dealing with fermions or bosons.[1] Thus, the wavefunction is defined for a fixed number of particles N, and it cannot describe processes that involve the creation or annihilation of particles.

This is a problem when we want to describe systems that can exchange particles with their environment, such as in the case of open systems or in the case of systems that can undergo particle creation and annihilation, such as in quantum field theory. To overcome this limitation, we need to introduce a new formalism, which is called **second quantization**, which allows us to describe systems with a variable number of particles by introducing creation and annihilation operators.

[1] The permanent of a matrix is the sum over all permutations of the matrix elements, with a plus sign for each permutation (while the determinant has alternating signs).

1.1 | Occupation Number Representation

In the context of second quantization, we can describe the state of a system of particles by specifying the **occupation numbers of the single-particle states** (seen as a set of quantum numbers). The occupation number n_ν of a single-particle state ν is defined as the number of particles that occupy that state.

Thus, we can represent the state of a N-particle system by a vector of occupation numbers:

$$|n_{\nu_1}, n_{\nu_2}, \dots, n_{\nu_m}\rangle, \quad \sum_\nu n_\nu = N,$$

where n_{ν_i} is the occupation number of the single-particle state ν_i, and the sum of all occupation numbers must be equal to the total number of particles N. This representation is called the **occupation number representation**, and it is a convenient way to describe many-body systems, since it allows us to keep track of the number of particles in each state without having to specify the coordinates of each particle. From this it follows straightforwardly the definition of the number operator $\hat{n}_\nu$, which counts the number of particles in the single-particle state ν:

$$\hat{n}_\nu |n_{\nu_1}, n_{\nu_2}, \dots, n_{\nu_m}\rangle = n_\nu |n_{\nu_1}, n_{\nu_2}, \dots, n_{\nu_m}\rangle .$$

For bosons, the occupation number can take any non-negative integer value, while for fermions, the occupation number can only be 0 or 1, due to the Pauli exclusion principle:

$$n_\nu = \begin{cases} 0, 1, 2, \dots & \text{for bosons,} \\ 0, 1 & \text{for fermions.} \end{cases}$$

In order to use this formalism, we need to introduce the analogue of the Hilbert space for the variable number of particles, which is called the **Fock space**. The Fock space is the direct sum of the Hilbert spaces for all possible numbers of particles:

$$\mathcal{F} = \bigoplus_{N=0}^{\infty} \mathcal{H}_N,$$

where $\mathcal{H}_N$ is the Hilbert space for N particles. The Fock space allows us to describe states with a variable number of particles, and it is the natural setting for the second quantization formalism. In the Fock space, we can define the vacuum state $|0\rangle$, which is the state with no particles, and we can build all other states by applying creation operators to the vacuum state. But before let us make an example.

> **Example (Fock space basis).** Let us consider a system of bosons with varying number of particles, and let us choose a single-particle basis with two states, which we can label as $|1\rangle$ and $|2\rangle$. The Fock space for this system is the direct sum of the Hilbert spaces for all possible numbers of particles:
>
> $$\mathcal{F} = \bigoplus_{N=0}^{\infty} \mathcal{H}_N.$$
>
> We can show some examples of basis for the Fock space:
>
> $$\begin{aligned} \mathcal{H}_0 &= |0, 0, \dots, 0\rangle , \\ \mathcal{H}_1 &= |1, 0, \dots, 0\rangle , |0, 1, \dots, 0\rangle , \dots , |0, 0, \dots, 1\rangle , \\ \mathcal{H}_2 &= |1, 1, 0, \dots, 0\rangle , \dots , |1, 0, \dots, 0, 1, 0, \dots, 0\rangle , |2, 0, \dots, 0\rangle , \dots \end{aligned}$$

where the first state is the vacuum state, the second states are the single-particle states, and the third states are the two-particle states. We can see that in the occupation number representation, we can easily keep track of the number of particles in each state, and we can build all possible states by permuting the occupation numbers.

Note that for a system of fermions, the basis states would be different, since the occupation numbers can only be 0 or 1, so we would not have states with more than one particle in the same state: the last state in the two-particle Hilbert space would not be allowed, since it would violate the Pauli exclusion principle.

Now it is time for us to introduce the creation and annihilation operators, which are the fundamental building blocks of the second quantization formalism.

1.2 | Ladder Operators

Ladder operators are operators that can raise or lower the occupation number of a single-particle state. We will separate the definition of the ladder operators for bosons and fermions, since they satisfy different commutation or anticommutation relations.

1.2.1 | Bosons

For bosons, we define the **creation operator** $\hat{b}_\nu^\dagger$ and the **annihilation operator** $\hat{b}_\nu$ for a single-particle state ν as follows:

$$\begin{cases} \hat{b}_\nu \left|n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu, \ldots\right\rangle = B_-(n_\nu) \left|n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu - 1, \ldots\right\rangle, \\ \hat{b}_\nu^\dagger \left|n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu, \ldots\right\rangle = B_+(n_\nu) \left|n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu + 1, \ldots\right\rangle, \end{cases}$$

where $B_-(n_\nu)$ and $B_+(n_\nu)$ are coefficients that depend on the occupation number n_ν. The creation operator $\hat{b}_\nu^\dagger$ increases the occupation number of the state ν by one, while the annihilation operator $\hat{b}_\nu$ decreases the occupation number of the state ν by one. For bosons, these operators satisfy the following commutation relations:

$$\begin{cases} [\hat{b}_\nu, \hat{b}_{\nu'}^\dagger] = \delta_{\nu\nu'}, \\ [\hat{b}_\nu, \hat{b}_{\nu'}] = [\hat{b}_\nu^\dagger, \hat{b}_{\nu'}^\dagger] = 0, \end{cases}$$

where $[A, B] = AB - BA$ is the commutator of the operators A and B. The commutation relations reflect the fact that bosons can occupy the same state, and that the order of the operators does not matter.

Let us apply these operators to the occupation number states to find the coefficients $B_-(n_\nu)$ and $B_+(n_\nu)$, starting from the vacuum state $\left|0, 0, \ldots, 0, \ldots\right\rangle$:

$$\begin{aligned} &\hat{b}_\nu \left|0, 0, \ldots, 0, \ldots\right\rangle = 0, && B_-(0) = 0, \\ &\hat{b}_\nu^\dagger \left|0, 0, \ldots, 0, \ldots\right\rangle = \left|0, 0, \ldots, 0, 1, 0, \ldots\right\rangle, && B_+(0) = 1, \end{aligned}$$

where we understood how the annihilation operator gives zero when applied to the vacuum state, while the creation operator creates a particle in the state ν.

Now let us apply the operators to a generic normalized occupation number state $\left|\phi\right\rangle$:

$$\hat{b}_\nu \left|\phi\right\rangle, \quad \left\langle\phi\right| \hat{b}_\nu^\dagger \hat{b}_\nu \left|\phi\right\rangle = |\hat{b}_\nu \left|\phi\right\rangle| = \lambda > 0,$$

such that if we apply the creation operator to the state $\hat{b}_\nu \left|\phi\right\rangle$ we get:

$$\hat{b}_\nu^\dagger \hat{b}_\nu \left|\phi\right\rangle = \lambda \left|\phi\right\rangle = n_\nu \left|\phi\right\rangle,$$

since the number operator $\hat{n}_\nu = \hat{b}_\nu^\dagger \hat{b}_\nu$ counts the number of particles in the state ν . Now if we apply the number operator $\hat{n}_\nu = \hat{b}_\nu^\dagger \hat{b}_\nu$ to the state $\hat{b}_\nu \left|\phi\right\rangle$, using the commutation relations, we should gbe able to find the coefficient $B_-(n_\nu)$:

$$\hat{n}_\nu \hat{b}_\nu \left|\phi\right\rangle = (\hat{b}_\nu^\dagger \hat{b}_\nu) \hat{b}_\nu \left|\phi\right\rangle = (\hat{b}_\nu \hat{b}_\nu^\dagger - 1) \hat{b}_\nu \left|\phi\right\rangle = \hat{b}_\nu (\hat{b}_\nu^\dagger \hat{b}_\nu - 1) \left|\phi\right\rangle = (\lambda - 1) \hat{b}_\nu \left|\phi\right\rangle,$$

so that we can identify the coefficient $B_-(n_\nu)$ as:

$$B_-(n_\nu) = \sqrt{n_\nu},$$

and by applying the number operator to the state $\hat{b}_\nu^\dagger |\phi\rangle$, we get:

$$\hat{n}_\nu \hat{b}_\nu^\dagger |\phi\rangle = (\hat{b}_\nu^\dagger \hat{b}_\nu)\hat{b}_\nu^\dagger |\phi\rangle = \hat{b}_\nu^\dagger (\hat{b}_\nu \hat{b}_\nu^\dagger) |\phi\rangle = \hat{b}_\nu^\dagger (\hat{b}_\nu^\dagger \hat{b}_\nu + 1) |\phi\rangle = (\lambda + 1)\hat{b}_\nu^\dagger |\phi\rangle ,$$

so that we can identify the coefficient $B_+(n_\nu)$ as:

$$B_+(n_\nu) = \sqrt{n_\nu + 1}.$$

Thus, the action of the creation and annihilation operators on the occupation number states can be expressed as:

$$\begin{cases} \hat{b}_\nu |n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu, \ldots\rangle = \sqrt{n_\nu}\, |n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu - 1, \ldots\rangle , \\ \hat{b}_\nu^\dagger |n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu, \ldots\rangle = \sqrt{n_\nu + 1}\, |n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu + 1, \ldots\rangle . \end{cases}$$

1.2.2 | Fermions

For fermions, we define the **creation operator** $\hat{c}_\nu^\dagger$ and the **annihilation operator** $\hat{c}_\nu$ for a single-particle state ν with the same action on the occupation number states as for bosons, but with the coefficients $B_-(n_\nu)$ and $B_+(n_\nu)$ that depend on the occupation number n_ν in a different way, since for fermions the occupation number can only be 0 or 1:

$$\begin{cases} \hat{c}_\nu |n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu, \ldots\rangle = C_-(n_\nu) |n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu - 1, \ldots\rangle , \\ \hat{c}_\nu^\dagger |n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu, \ldots\rangle = C_+(n_\nu) |n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu + 1, \ldots\rangle , \end{cases}$$

with

$$C_-(n_\nu) = \begin{cases} 0 & \text{if } n_\nu = 0, \\ 1 & \text{if } n_\nu = 1, \end{cases} \quad C_+(n_\nu) = \begin{cases} 1 & \text{if } n_\nu = 0, \\ 0 & \text{if } n_\nu = 1, \end{cases}$$

so that the creation operator $\hat{c}_\nu^\dagger$ creates a particle in the state ν if it is not already occupied, while the annihilation operator $\hat{c}_\nu$ annihilates a particle in the state ν if it is occupied. For fermions, these operators satisfy the following anticommutation relations:

$$\begin{cases} \{\hat{c}_\nu, \hat{c}_\nu^\dagger\} = 1, \\ \{\hat{c}_\nu, \hat{c}_\nu\} = \{\hat{c}_\nu^\dagger, \hat{c}_\nu^\dagger\} = 0, \end{cases}$$

since the multiparticle wavefunction must be antisymmetric under the exchange of any two particles, and thus the order of the operators matters. The anticommutation relations reflect the fact that fermions cannot occupy the same state:

$$\hat{c}_\nu^\dagger \hat{c}_\nu^\dagger = \frac{1}{2}\{\hat{c}_\nu^\dagger, \hat{c}_\nu^\dagger\} = 0,$$

and similarly for the annihilation operator.

1.3 | Hamiltonian of an Interacting System

In a many-body system, the Hamiltonian has to contain both a kinetic term, which describes the motion of the particles, and an interaction term, which describes the interactions between the particles. Let us show how to write the Hamiltonian of an interacting system in the second quantization formalism, starting from the first quantization formalism. We will see how many results from first quantization come in handy when it will be time to build the second quantization version.

First Quantization

In first quantization, the Hamiltonian of a system of N interacting particles can be written as:

$$H = \sum_{i=1}^{N} \hat{T}(\mathbf{r}_i, \nabla_{\mathbf{r}_i}) + \frac{1}{2}\sum_{i\neq j} \hat{V}(\mathbf{r}_i - \mathbf{r}_j),$$

where $\hat{T}(\mathbf{r}_i, \nabla_{\mathbf{r}_i})$ is the **one-body operator** for the i-th particle, and $\hat{V}(\mathbf{r}_i - \mathbf{r}_j)$ is the interaction potential, or **two body operator**, between the i-th and j-th particles. The factor of $\frac{1}{2}$ is included to avoid double counting the interactions between pairs of particles.

The easiest case in which we can write the Hamiltonian of a many-body system is the Helium atom.

> **Example (Helium atom).** Let us consider the helium atom, which consists of two electrons interacting with each other and with the nucleus. The Hamiltonian of the helium atom can be written as:
>
> $$H = \left[-\frac{\hbar^2}{2m}\nabla^2_{\mathbf{r}_1} - \frac{Ze^2}{4\pi\epsilon_0 r_1}\right] + \left[-\frac{\hbar^2}{2m}\nabla^2_{\mathbf{r}_2} - \frac{Ze^2}{4\pi\epsilon_0 r_2}\right] + \frac{e^2}{4\pi\epsilon_0|\mathbf{r}_1 - \mathbf{r}_2|},$$
>
> where m is the mass of the electron, Z is the atomic number of the nucleus (which is 2 for helium), e is the elementary charge, ϵ_0 is the vacuum permittivity and r_i is the distance of the i-th electron from the nucleus. The first term in square brackets represent the contribution of the kinetic energy and the potential energy of the first electron, while the second term in square brackets for the second electron; the last term represent the interaction between the two electrons, which is the **electron-electron interaction**. This Hamiltonian is a good starting point for understanding the electronic structure of the helium atom, and it can be solved using various approximation methods, such as perturbation theory or variational methods.

Let us study more in detail the structure of the one-body and two-body operators.

One-body operator. The one-body operator $\hat{T}(\mathbf{r}_j, \nabla_{\mathbf{r}_j})$ is an operator that acts on the coordinates of a single particle, and it typically includes the kinetic energy term and the potential energy term due to external fields. It can be expressed in terms of the single-particle states as:

$$T_{\nu_a\nu_b} = \int \mathrm{d}^3\mathbf{r}_j\, \Psi^*_{\nu_a}(\mathbf{r}_j)\hat{T}(\mathbf{r}_j, \nabla_{\mathbf{r}_j})\Psi_{\nu_b}(\mathbf{r}_j),$$

where $\Psi_\nu(\mathbf{r}_j)$ are the single-particle wavefunctions, and the integral is taken over all space. The internal operator $\hat{T}(\mathbf{r}_j, \nabla_{\mathbf{r}_j})$ can be expressed as the sum of the kinetic energy operator and the potential energy operator:

$$\hat{T}(\mathbf{r}_j, \nabla_{\mathbf{r}_j}) = -\frac{\hbar^2}{2m}\nabla^2_{\mathbf{r}_j} + V_{\text{ext}}(\mathbf{r}_j),$$

and since $T_{\nu_a\nu_b}$ is the matrix element of the one-body operator between the single-particle states ν_a and ν_b, we can write:

$$\hat{T}(\mathbf{r}_j, \nabla_{\mathbf{r}_j}) = \sum_{\nu_a\nu_b} T_{\nu_a\nu_b} \left|\Psi_{\nu_b}(\mathbf{r}_j)\right\rangle \left\langle\Psi_{\nu_a}(\mathbf{r}_j)\right|,$$

where $|\Psi_{\nu_a}(\mathbf{r}_j)\rangle$ and $|\Psi_{\nu_b}(\mathbf{r}_j)\rangle$ are the single-particle states corresponding to the quantum numbers ν_a and ν_b, respectively.

Two-body operator. The two-body operator $\hat{V}(\mathbf{r}_j - \mathbf{r}_k)$ is an operator that acts on the coordinates of two particles, and it typically includes the interaction potential between the particles. It can be expressed in terms of the single-particle states as:

$$V_{\nu_c\nu_d,\nu_a\nu_b} = \int \mathrm{d}^3\mathbf{r}_j\, \mathrm{d}^3\mathbf{r}_k\, \Psi^*_{\nu_c}(\mathbf{r}_j)\Psi^*_{\nu_d}(\mathbf{r}_k)\hat{V}(\mathbf{r}_j - \mathbf{r}_k)\Psi_{\nu_a}(\mathbf{r}_j)\Psi_{\nu_b}(\mathbf{r}_k),$$

where again $\Psi_\nu(\mathbf{r}_j)$ are the single-particle wavefunctions, and the integral is taken over all space. The internal operator $\hat{V}(\mathbf{r}_j - \mathbf{r}_k)$ can then be expressed as a sum over the single-particle states as:

$$\hat{V}(\mathbf{r}_j - \mathbf{r}_k) = \sum_{\nu_a\nu_b\nu_c\nu_d} V_{\nu_c\nu_d,\nu_a\nu_b} \left|\Psi_{\nu_c}(\mathbf{r}_j)\right\rangle \left|\Psi_{\nu_d}(\mathbf{r}_k)\right\rangle \left\langle\Psi_{\nu_b}(\mathbf{r}_k)\right| \left\langle\Psi_{\nu_a}(\mathbf{r}_j)\right|,$$

where $V_{\nu_c\nu_d,\nu_a\nu_b}$ are the matrix elements of the two-body operator between the single-particle states.

Now that we have expressed the one-body and two-body operators in terms of the single-particle states, we can rewrite the Hamiltonian of the system in terms of these operators as:

$$H = \sum_{i=1}^{N}\sum_{\nu_a\nu_b} T_{\nu_a\nu_b} \left|\Psi_{\nu_b}(\mathbf{r}_i)\right\rangle \left\langle\Psi_{\nu_a}(\mathbf{r}_i)\right| + \frac{1}{2}\sum_{i\neq j}\sum_{\nu_a\nu_b\nu_c\nu_d} V_{\nu_c\nu_d,\nu_a\nu_b} \left|\Psi_{\nu_c}(\mathbf{r}_i)\right\rangle \left|\Psi_{\nu_d}(\mathbf{r}_j)\right\rangle \left\langle\Psi_{\nu_b}(\mathbf{r}_j)\right| \left\langle\Psi_{\nu_a}(\mathbf{r}_i)\right|,$$

where we have expressed the Hamiltonian in terms of the matrix elements of the one-body and two-body operators between the single-particle states. This Hamiltonian is still in a first quantization form, since it is expressed in terms of the coordinates of the particles, but it is a convenient starting point for expressing the Hamiltonian in terms of the creation and annihilation operators in the second quantization formalism.

Second Quantization

In order to express the Hamiltonian in terms of the creation and annihilation operators, we need to understand how to express the operators in second quantization. The key idea is to express the operators in terms of the ladder operators, which can be done by using the completeness relation of the single-particle states.

It can be shown that an operator in first quantization can be expressed in second quantization as:

$$\hat{A} = \sum_{\nu_a\nu_b} A_{\nu_a\nu_b}\hat{a}^\dagger_{\nu_a}\hat{a}_{\nu_b},$$

where $A_{\nu_a\nu_b}$ are the matrix elements of the operator in the single-particle basis, and $\hat{a}^\dagger_\nu$ and $\hat{a}_\nu$ are the creation and annihilation operators for the single-particle state ν, respectively.[2] Thus there exists a systematic way to express any operator in first quantization in terms of the creation

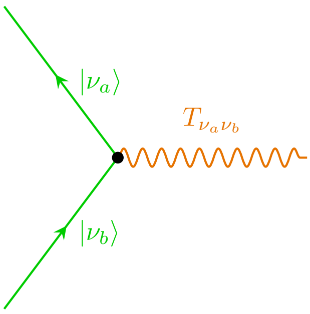
[2] To be substitued by $\hat{b}$ and $\hat{c}$ according to the statistics of the particles.

and annihilation operators in second quantization, by expressing the operator in terms of its matrix elements in the single-particle basis, and then replacing the single-particle states with the corresponding creation and annihilation operators.

We can then write the Hamiltonian of the system in second quantization as:

$$H = \sum_{\nu_a \nu_b} T_{\nu_a \nu_b} \hat{a}^\dagger_{\nu_a} \hat{a}_{\nu_b} + \frac{1}{2} \sum_{\nu_a \nu_b \nu_c \nu_d} V_{\nu_c \nu_d, \nu_a \nu_b} \hat{a}^\dagger_{\nu_c} \hat{a}^\dagger_{\nu_d} \hat{a}_{\nu_a} \hat{a}_{\nu_b},$$

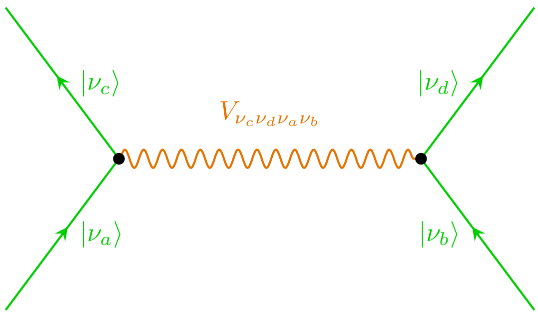
where the first term represents the one-body interactions, and the second term represents the two-body interactions between the particles. This Hamiltonian is now expressed in terms of the creation and annihilation operators, and it can be used to describe systems with a variable number of particles, since the creation and annihilation operators can create or annihilate particles in the system.



In the diagram on the left, we can see the one-body interaction between the single-particle states $|\nu_a\rangle$ and $|\nu_b\rangle$, which is represented by the matrix element $T_{\nu_a \nu_b}$. The particle represented by the line entering the vertex is in the state $|\nu_b\rangle$, while the particle represented by the line exiting the vertex is in the state $|\nu_a\rangle$. The interaction is represented by a wavy line, which indicates that it is a one-body interaction, since it involves only one particle.

The vertex represents the point where the interaction takes place, and it is associated with the matrix element $T_{\nu_a \nu_b}$, which quantifies the strength of the interaction between the two states, or else the probability amplitude for the transition from state $|\nu_b\rangle$ to state $|\nu_a\rangle$ due to the one-body operator. Basically we are annihilating a particle in the state $|\nu_b\rangle$ and creating a particle in the state $|\nu_a\rangle$ due to the one-body interaction.

For the two body interaction, represented in the diagram on the right, we have two vertices representing the interaction between the single-particle states $|\nu_a\rangle$, $|\nu_b\rangle$, $|\nu_c\rangle$ and $|\nu_d\rangle$. The particles represented by the lines entering the vertices are in the states $|\nu_a\rangle$ and $|\nu_b\rangle$, while the particles represented by the lines exiting the vertices are in the states $|\nu_c\rangle$ and $|\nu_d\rangle$.



The interaction is represented by a wavy line connecting the two vertices, which indicates that it is a two-body interaction, since it involves two particles. The matrix element $V_{\nu_c \nu_d \nu_a \nu_b}$ quantifies the strength of the interaction between the four states, or else the probability amplitude for the transition from states $|\nu_a\rangle$ and $|\nu_b\rangle$ to states $|\nu_c\rangle$ and $|\nu_d\rangle$ due to the two-body operator. We are describing a process in which we are annihilating two particles in the states $|\nu_a\rangle$ and $|\nu_b\rangle$ and creating two particles in the states $|\nu_c\rangle$ and $|\nu_d\rangle$ due to the two-body interaction.

1.4 | Field Operators

Field operators are a powerful tool in quantum mechanics, particularly in the study of many-body systems. They allow us to describe the creation and annihilation of particles in a given state. The field operator $\hat{\Psi}(\mathbf{r})$ can be expressed as a sum over a complete set of single-particle states:

$$\hat{\Psi}(\mathbf{r}) = \sum_{\nu} \phi_{\nu}(\mathbf{r})\hat{a}_{\nu},$$
$$\hat{\Psi}^{\dagger}(\mathbf{r}) = \sum_{\nu} \phi^{*}_{\nu}(\mathbf{r})\hat{a}^{\dagger}_{\nu},$$

where $\phi_{\nu}(\mathbf{r})$ are the single-particle wavefunctions on the phase space, and $\hat{a}_{\nu}$ and $\hat{a}^{\dagger}_{\nu}$ are the annihilation and creation operators for the single-particle state ν, respectively.[3] The field operator $\hat{\Psi}(\mathbf{r})$ annihilates a particle at position $\mathbf{r}$, while the field operator $\hat{\Psi}^{\dagger}(\mathbf{r})$ creates a particle at position $\mathbf{r}$; they are defined in such a way that they satisfy the appropriate commutation or anticommutation relations, depending on whether we are dealing with bosons or fermions.

If we have an **homogeneous system**, we can choose plane waves as our single-particle states to sum only over momentum states, moving to the **momentum space** representation. In this case, the field operators can be expressed as:

$$\hat{\Psi}(\mathbf{r}) = \frac{1}{\sqrt{\Omega}} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}}\hat{a}_{\mathbf{k}},$$
$$\hat{\Psi}^{\dagger}(\mathbf{r}) = \frac{1}{\sqrt{\Omega}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}}\hat{a}^{\dagger}_{\mathbf{k}},$$

we are summing over all quantum numbers ν, which in this case are the momentum states $\mathbf{k}$, then we are creating or annihilating particles in all of those momentum states for a precise position $\mathbf{r}$. Since this operators are based on the same annihilation and creation operators, they satisfy the same commutation or anticommutation relations, depending on whether we are dealing with bosons or fermions. For bosons, the field operators satisfy the commutation relations:

$$\begin{cases} [\hat{\Psi}(\mathbf{r}), \hat{\Psi}^{\dagger}(\mathbf{r}')] = \delta(\mathbf{r} - \mathbf{r}'), \\ [\hat{\Psi}(\mathbf{r}), \hat{\Psi}(\mathbf{r}')] = [\hat{\Psi}^{\dagger}(\mathbf{r}), \hat{\Psi}^{\dagger}(\mathbf{r}')] = 0, \end{cases}$$

while for fermions, they satisfy the anticommutation relations:

$$\begin{cases} \{\hat{\Psi}(\mathbf{r}), \hat{\Psi}^{\dagger}(\mathbf{r}')\} = \delta(\mathbf{r} - \mathbf{r}'), \\ \{\hat{\Psi}(\mathbf{r}), \hat{\Psi}(\mathbf{r}')\} = \{\hat{\Psi}^{\dagger}(\mathbf{r}), \hat{\Psi}^{\dagger}(\mathbf{r}')\} = 0, \end{cases}$$

derived from the commutation or anticommutation relations of the ladder operators. Instead of wavefunctions, we can also express the field operators in terms of the position basis.

Interacting Hamiltonian in Terms of Field Operators

Let us now express the Hamiltonian of an interacting system in terms of the field operators. Starting from the Hamiltonian in second quantization, derived in the previous section, it can be expressed as:

$$\hat{H} = \sum_{\nu_a, \nu_b} \hat{T}_{\nu_a \nu_b} \hat{a}^{\dagger}_{\nu_a} \hat{a}_{\nu_b} + \frac{1}{2} \sum_{\nu_a, \nu_b, \nu_c, \nu_d} \hat{V}_{\nu_c \nu_d \nu_a \nu_b} \hat{a}^{\dagger}_{\nu_c} \hat{a}^{\dagger}_{\nu_d} \hat{a}_{\nu_a} \hat{a}_{\nu_b},$$

[3] Here we use generic ladder operators $\hat{a}$, but according to the context, we might use $\hat{c}$ for electrons or $\hat{b}$ for bosons, along with the respective commutation relations.

where we are summing over all quantum numbers ν. Now rewriting it in terms of the field operators is a straightforward task, since we can express the one-body and two-body operators in a more explicit way as:

$$\hat{H} = \sum_{\nu_a, \nu_b} \int \mathrm{d}^3\mathbf{r} \, \left(\Psi^*_{\nu_a}(\mathbf{r}) T(\mathbf{r}, \nabla_{\mathbf{r}}) \Psi_{\nu_b}(\mathbf{r})\right) \hat{a}^\dagger_{\nu_a} \hat{a}_{\nu_b}$$
$$+ \frac{1}{2} \sum_{\nu_a, \nu_b, \nu_c, \nu_d} \left(\int \mathrm{d}^3\mathbf{r} \, \mathrm{d}^3\mathbf{r}' \, \Psi^*_{\nu_c}(\mathbf{r}) \Psi^*_{\nu_d}(\mathbf{r}') V(\mathbf{r} - \mathbf{r}') \Psi_{\nu_a}(\mathbf{r}) \Psi_{\nu_b}(\mathbf{r}') \right) \hat{a}^\dagger_{\nu_c} \hat{a}^\dagger_{\nu_d} \hat{a}_{\nu_a} \hat{a}_{\nu_b},$$

from which is now straightforward to recognize the expressions of the field operators (with $\Psi_\nu(\mathbf{r})$ the single-particle wavefunctions), so that we can rewrite the Hamiltonian as:

$$\hat{H} = \int \mathrm{d}^3\mathbf{r} \, \hat{\Psi}^\dagger(\mathbf{r}) T(\mathbf{r}, \nabla_{\mathbf{r}}) \hat{\Psi}(\mathbf{r}) + \frac{1}{2} \int \mathrm{d}^3\mathbf{r} \, \mathrm{d}^3\mathbf{r}' \, \hat{\Psi}^\dagger(\mathbf{r}) \hat{\Psi}^\dagger(\mathbf{r}') V(\mathbf{r} - \mathbf{r}') \hat{\Psi}(\mathbf{r}') \hat{\Psi}(\mathbf{r}),$$

where the first term represents the kinetic energy and the potential energy due to external fields, while the second term represents the interaction energy between the particles. This Hamiltonian is now expressed in terms of the field operators, and it can be used to describe systems with a variable number of particles, since the field operators can create or annihilate particles in a specific state in the system.

Let us now analyze the case in which the one-body operator is given by the kinetic term, and the two-body operator is given by the Coulomb potential, which is a common case in many-body physics.

Free kinetic term. The kinetic term of the Hamiltonian can be expressed as:

- in **first quantization**, the kinetic term is given by:

  $$\hat{T}(\mathbf{r}, \nabla_{\mathbf{r}}) = -\frac{\hbar^2}{2m} \nabla^2_{\mathbf{r}} \delta_{\sigma\sigma'},$$

  where we are using the plane wave basis $|\mathbf{k}, \sigma\rangle$, such that

  $$\langle \mathbf{r} | \mathbf{k}, \sigma \rangle = \frac{1}{\sqrt{\Omega}} e^{i\mathbf{k}\cdot\mathbf{r}},$$

  and if we compute the matrix element of the kinetic term, we get:

  $$\langle \mathbf{k}_a, \sigma_a | \, \hat{T}(\mathbf{r}, \nabla_{\mathbf{r}}) \, | \mathbf{k}_b, \sigma_b \rangle = \frac{\hbar^2 k_b^2}{2m} \delta_{\mathbf{k}_a \mathbf{k}_b} \delta_{\sigma_a \sigma_b},$$

  since the plane waves are eigenstates of the kinetic energy operator (the laplacian is diagonal), and the eigenvalue is given by $\frac{\hbar^2 k^2}{2m}$. The delta functions ensure that the kinetic term only contributes when the momentum and spin states are the same.

- in **second quantization**, the kinetic term can be expressed as:

  $$\hat{T} = \sum_{\nu_a, \nu_b} T_{\nu_a \nu_b} \hat{a}^\dagger_{\nu_a} \hat{a}_{\nu_b} = \sum_{\mathbf{k}, \mathbf{k}', \sigma, \sigma'} \langle \mathbf{k}, \sigma | \, \hat{T}(\mathbf{r}, \nabla_{\mathbf{r}}) \, | \mathbf{k}', \sigma' \rangle \, \hat{a}^\dagger_{\mathbf{k},\sigma} \hat{a}_{\mathbf{k}',\sigma'} = \sum_{\mathbf{k},\sigma} \frac{\hbar^2 k^2}{2m} \hat{a}^\dagger_{\mathbf{k},\sigma} \hat{a}_{\mathbf{k},\sigma},$$

  where we have used the fact that the kinetic term is diagonal in the plane wave basis, so that the only non-zero contributions come from terms where $\mathbf{k} = \mathbf{k}'$ and $\sigma = \sigma'$. Indeed, we can see that in the last term the kinetic term is expressed as a sum over the momentum states, with the eigenvalue of the kinetic energy operator as the coefficient, and the number operator for the corresponding momentum states:

  $$\hat{T} = \sum_{\mathbf{k},\sigma} \frac{\hbar^2 k^2}{2m} \hat{n}_{\mathbf{k},\sigma}.$$

Coulomb interaction term. Starting directly from the second quantized expression of the Coulomb term of the Hamiltonian, we can express it as:

$$V = \frac{1}{2}\sum_{\sigma,\sigma'}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} \langle \mathbf{k}_3, \sigma; \mathbf{k}_4, \sigma' | \, V(\mathbf{r}-\mathbf{r}') \, | \mathbf{k}_2, \sigma'; \mathbf{k}_1, \sigma\rangle \, \hat{a}^\dagger_{\mathbf{k}_3,\sigma}\hat{a}^\dagger_{\mathbf{k}_4,\sigma'}\hat{a}_{\mathbf{k}_2,\sigma'}\hat{a}_{\mathbf{k}_1,\sigma},$$

where we are summing over all momentum and spin states, and the matrix element of the Coulomb potential can be expressed using the normalized plane wave basis as:

$$V = \frac{1}{2}\sum_{\sigma,\sigma'}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4}\left[\int\int \mathrm{d}^3\mathbf{r}\,\mathrm{d}^3\mathbf{r}'\,\frac{e^{-i\mathbf{k}_3\cdot\mathbf{r}}}{\sqrt{\Omega}}\frac{e^{-i\mathbf{k}_4\cdot\mathbf{r}'}}{\sqrt{\Omega}}V(\mathbf{r}-\mathbf{r}')\frac{e^{i\mathbf{k}_2\cdot\mathbf{r}'}}{\sqrt{\Omega}}\frac{e^{i\mathbf{k}_1\cdot\mathbf{r}}}{\sqrt{\Omega}}\right]\hat{a}^\dagger_{\mathbf{k}_3,\sigma}\hat{a}^\dagger_{\mathbf{k}_4,\sigma'}\hat{a}_{\mathbf{k}_2,\sigma'}\hat{a}_{\mathbf{k}_1,\sigma},$$

and now, exploiting the form of the Coulomb potential, we can express the matrix element as:

$$V = \frac{1}{2}\sum_{\sigma,\sigma'}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4}\frac{e^2}{\Omega^2}\left[\int\int \mathrm{d}^3\mathbf{r}\,\mathrm{d}^3\mathbf{r}'\,\frac{e^{i(\mathbf{k}_1-\mathbf{k}_3)\cdot\mathbf{r}}e^{i(\mathbf{k}_2-\mathbf{k}_4)\cdot\mathbf{r}'}}{|\mathbf{r}-\mathbf{r}'|}\right]\hat{a}^\dagger_{\mathbf{k}_3,\sigma}\hat{a}^\dagger_{\mathbf{k}_4,\sigma'}\hat{a}_{\mathbf{k}_2,\sigma'}\hat{a}_{\mathbf{k}_1,\sigma}.$$

We can then perform the change of variables $\boldsymbol{\xi} = \mathbf{r}-\mathbf{r}'$, so that the integral can be expressed as:

$$V = \frac{1}{2}\sum_{\sigma,\sigma'}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4}\frac{e^2}{\Omega^2}\left[\int \mathrm{d}^3\mathbf{r}\, e^{i(\mathbf{k}_1-\mathbf{k}_3+\mathbf{k}_2-\mathbf{k}_4)\cdot\mathbf{r}}\int \mathrm{d}^3\boldsymbol{\xi}\, e^{i(\mathbf{k}_2-\mathbf{k}_4)\cdot(\mathbf{r}-\boldsymbol{\xi})}\frac{1}{|\boldsymbol{\xi}|}\right]\hat{a}^\dagger_{\mathbf{k}_3,\sigma}\hat{a}^\dagger_{\mathbf{k}_4,\sigma'}\hat{a}_{\mathbf{k}_2,\sigma'}\hat{a}_{\mathbf{k}_1,\sigma},$$

where we have separated the integral over $\mathbf{r}$ and the integral over $\boldsymbol{\xi}$. Now the equation is basically telling us to rename $\mathbf{k}_2 - \mathbf{k}_4$ as $\mathbf{q}$:

$$V = \frac{1}{2}\sum_{\sigma,\sigma'}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{q}}\frac{e^2}{\Omega^2}\left[\int \mathrm{d}^3\mathbf{r}\, e^{i(\mathbf{k}_1-\mathbf{k}_3+\mathbf{q})\cdot\mathbf{r}}\int \mathrm{d}^3\boldsymbol{\xi}\, e^{i\mathbf{q}\cdot(\mathbf{r}-\boldsymbol{\xi})}\frac{1}{|\boldsymbol{\xi}|}\right]\hat{a}^\dagger_{\mathbf{k}_3,\sigma}\hat{a}^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma'}\hat{a}_{\mathbf{k}_2,\sigma'}\hat{a}_{\mathbf{k}_1,\sigma},$$

where we have also relabeled the indices of the creation and annihilation operators according to the change of variables. In the end we found the *Fourier transform of the Coulomb potential*:

$$\tilde{V} = \int \mathrm{d}^3\boldsymbol{\xi}\,\frac{e^{i\mathbf{q}\cdot\boldsymbol{\xi}}}{|\boldsymbol{\xi}|} = \frac{4\pi}{|\mathbf{q}|^2}.$$

As we can see, the Coulomb potential is a very **long range potential**, since its transform as $\mathbf{q}\to 0$ diverges:

$$\lim_{\mathbf{q}\to 0}\tilde{V} = \frac{4\pi}{|\mathbf{q}|^2},$$

since it is not screened by the presence of other particles in the system, and it can have a significant effect on its properties, even at large distances. In the end, integrating over $\mathbf{r}$ we get

$$V = \frac{1}{2}\sum_{\sigma,\sigma'}\sum_{\mathbf{k}_1,\mathbf{k}_3,\mathbf{q}}\frac{e^2}{\Omega^2}\Omega\,\delta_{\mathbf{k}_3,\mathbf{k}_1+\mathbf{q}}\,\hat{a}^\dagger_{\mathbf{k}_3,\sigma}\hat{a}^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma'}\hat{a}_{\mathbf{k}_2,\sigma'}\hat{a}_{\mathbf{k}_1,\sigma},$$

where the delta function enforces the **conservation of momentum** during the interaction, so that the total momentum before and after the interaction is the same. Finally, we can sum over the delta function, so that we can express the Coulomb term as:

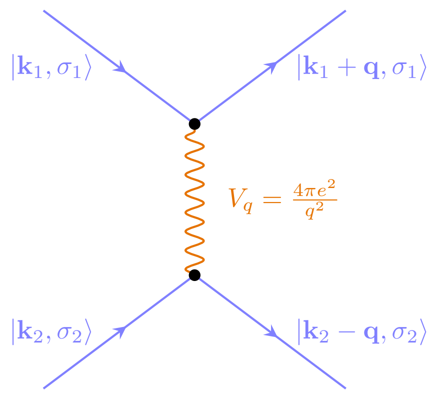
$$V = \frac{1}{2\Omega}\sum_{\sigma,\sigma'}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{q}}\frac{4\pi e^2}{|\mathbf{q}|^2}\hat{a}^\dagger_{\mathbf{k}_1+\mathbf{q},\sigma}\hat{a}^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma'}\hat{a}_{\mathbf{k}_2,\sigma'}\hat{a}_{\mathbf{k}_1,\sigma}.$$

Thus during the interaction, two particles with momenta $\mathbf{k}_1$ and $\mathbf{k}_2$ exchange a momentum $\mathbf{q}$ and end up with momenta $\mathbf{k}_1+\mathbf{q}$ and $\mathbf{k}_2-\mathbf{q}$, so that the total momentum is conserved during the interaction.[4] This is a consequence of the translational invariance of the system, which implies that the total momentum is a good quantum number:

$$\mathbf{k}_3 + \mathbf{k}_4 = (\mathbf{k}_1 + \mathbf{q}) + (\mathbf{k}_2 - \mathbf{q}) = \mathbf{k}_1 + \mathbf{k}_2.$$

For the spin part, we know that the *Coulomb interaction does not depend on the spin of the particles*,[5] so that the spin is also conserved during the interaction, which is a consequence of the rotational invariance of the system.

Thus we have modeled the interaction between the particles as a two-body interaction, which then scatters two particles with momenta $\mathbf{k}_1$ and $\mathbf{k}_2$ into two particles with momenta $\mathbf{k}_1 + \mathbf{q}$ and $\mathbf{k}_2 - \mathbf{q}$, where the momentum $\mathbf{q}$ is the momentum exchanged during the interaction. This is a very general form of the interaction, which can be applied to any system with translational invariance, and it is the starting point for many-body perturbation theory and other techniques used to study interacting systems.



This Hamiltonian cannot be solved exactly (for more than two particles), so we need to use some approximation methods to study the properties of the system. We are so going to develop the many body perturbation theory in the next chapters, which is a powerful tool to study interacting systems, and it is based on the idea of treating the interaction as a perturbation to the non-interacting system.

[4]The momentum conservation is enforced at the end by the delta finction $\delta_{\mathbf{k}_1, \mathbf{k}_3 + \mathbf{q}}$.
[5]In some kind of interactions, we have other terms including all the Pauli matrices, so spin can switch.

2 The Electron Gas

The electron gas is a fundamental model in condensed matter physics that describes a system of free electrons: it is a widely used model to study different kinds of materials and systems. The electron gas is indeed a useful tool to understand the behavior of electrons in a solid, and it captures many of the essential features of real materials, such as the Fermi surface, the density of states, and the response functions.

2.1 | Jellium Model

The electron gas is a model of a system of electrons that are free to move in a uniform background of positive charge, which is often used to study the properties of metals and other materials. This model is also known as the **jellium model**, and it is a useful tool to understand the behavior of electrons in a solid. Its main features are:

- The electrons are treated as *free particles*, which means that they do not interact with each other, but they do interact with the positive background charge.
- The positive background charge is treated as a *uniform distribution of charge*, which means that it does not have any structure or dynamics.
- The system is assumed to be *infinite and homogeneous*, which means that it has no boundaries and that the properties of the system are the same everywhere.

The Hamiltonian of the electron gas can be expressed as:

$$H = (T_{\text{ion}} + V_{\text{ion-ion}}) + (T_{\text{el}} + V_{\text{el-el}}) + V_{\text{el-ion}},$$

where T_{ion} is the kinetic energy of the ions, $V_{\text{ion-ion}}$ is the potential energy of the ion-ion interaction, T_{el} is the kinetic energy of the electrons, $V_{\text{el-el}}$ is the potential energy of the electron-electron interaction, and $V_{\text{el-ion}}$ is the potential energy of the electron-ion interaction.

2.1.1 | Jellium Hamiltonian

Let us review term by term, starting from the ionic part:[1]

$$\begin{aligned} T_{\text{ion}} + V_{\text{ion-ion}} &= \int \mathrm{d}^3\mathbf{R}\hat{\Psi}^{\dagger}_{\text{ion}}(\mathbf{R}) \left(-\frac{\hbar^2}{2M}\nabla^2_{\mathbf{R}}\right) \hat{\Psi}_{\text{ion}}(\mathbf{R}) \\ +\frac{1}{2}\int\int \mathrm{d}^3\mathbf{R}_1\,\mathrm{d}^3\mathbf{R}_2\,\hat{\Psi}^{\dagger}_{\text{ion}}(\mathbf{R}_1)\hat{\Psi}^{\dagger}_{\text{ion}}(\mathbf{R}_2)&\frac{Z^2e^2}{|\mathbf{R}_1-\mathbf{R}_2|}\hat{\Psi}_{\text{ion}}(\mathbf{R}_2)\hat{\Psi}_{\text{ion}}(\mathbf{R}_1), \end{aligned}$$

while for the electrons we have:

$$\begin{aligned} T_{\text{el}} + V_{\text{el-el}} &= \int \mathrm{d}^3\mathbf{r}\hat{\Psi}^{\dagger}_{\text{el}}(\mathbf{r}) \left(-\frac{\hbar^2}{2m}\nabla^2_{\mathbf{r}}\right) \hat{\Psi}_{\text{el}}(\mathbf{r}) \\ +\frac{1}{2}\int\int \mathrm{d}^3\mathbf{r}_1\,\mathrm{d}^3\mathbf{r}_2\,\hat{\Psi}^{\dagger}_{\text{el}}(\mathbf{r}_1)\hat{\Psi}^{\dagger}_{\text{el}}(\mathbf{r}_2)&\frac{e^2}{|\mathbf{r}_1-\mathbf{r}_2|}\hat{\Psi}_{\text{el}}(\mathbf{r}_2)\hat{\Psi}_{\text{el}}(\mathbf{r}_1). \end{aligned}$$

The remaining term is the electron-ion interaction, which can be expressed as:

$$V_{\text{el-ion}} = \sum_{\sigma}\int\int \mathrm{d}^3\mathbf{r}\,\mathrm{d}^3\mathbf{R}\,\hat{\Psi}^{\dagger}_{\text{el},\sigma}(\mathbf{r})\hat{\Psi}^{\dagger}_{\text{ion}}(\mathbf{R}) \left(\frac{-Ze^2}{|\mathbf{r}-\mathbf{R}|}\right) \hat{\Psi}_{\text{ion}}(\mathbf{R})\hat{\Psi}_{\text{el},\sigma}(\mathbf{r}).$$

In the jellium model, we assume that the ions are fixed in space and that they form a **uniform background of positive charge**, while the electrons are free to move in this background, such that the density can be expressed as:

$$\rho_{jel}(\mathbf{R}) = \frac{N}{\Omega},$$

where N is the number of nuclei and Ω is the volume of the system, such is constant. In the end the idea is to get rid of the ionic part (the nuclei, which have constant density) and to *focus only on*

[1] Capital letters for nuclei coordinates and small letters for electrons coordinates.

the electronic part, which is the one that determines the properties of the system. Indeed, making this assumption, we can see that the complicated *periodic potential* of the ions is replaced by a simple *uniform background*.

Ion-ion interaction. We can use the jellium assumption to simplify the ion-ion interaction, which can be expressed as

$$V_{\text{ion-ion}} = \frac{1}{2} \int \int \mathrm{d}^3\mathbf{R}_1 \, \mathrm{d}^3\mathbf{R}_2 \, \rho_{jel}(\mathbf{R}_1)\rho_{jel}(\mathbf{R}_2) \frac{1}{|\mathbf{R}_1 - \mathbf{R}_2|},$$

where we are basically multiplicating the two infinitesimal charges $\rho_{jel}(\mathbf{R}_1)\mathrm{d}^3\mathbf{R}_1$ and $\rho_{jel}(\mathbf{R}_2)\mathrm{d}^3\mathbf{R}_2$ and then integrating over all space after dividing by the distance between the two charges. Thus we can write

$$\begin{aligned} V_{\text{ion-ion}} &= \frac{1}{2} \int \int \mathrm{d}^3\mathbf{R}_1 \, \mathrm{d}^3\mathbf{R}_2 \, \frac{N^2}{\Omega^2} \frac{1}{|\mathbf{R}_1 - \mathbf{R}_2|} \\ &= \frac{N^2}{2\Omega^2} \int \mathrm{d}^3\mathbf{R} \int \mathrm{d}^3\boldsymbol{\xi} \, \frac{1}{|\boldsymbol{\xi}|} = \frac{N^2}{2\Omega^2} \, \Omega \lim_{q \to 0} V_q, \end{aligned}$$

where in the end we have performed the change of variables $\boldsymbol{\xi} = \mathbf{R}_1 - \mathbf{R}_2$ and we have expressed the integral in terms of the Fourier transform of the Coulomb potential, which diverges as $q \to 0$. This means that the ion-ion interaction diverges in the thermodynamic limit, which is a problem for the theory, but we will see that this divergence is cancelled by the electron-ion interaction, so that the final Hamiltonian of the electron gas is well defined and does not diverge.

Electron-ion interaction. Moving on to the electron-ion interaction (among electrons and the jellium background), we can express it as:

$$\begin{aligned} V_{\text{el-ion}} &= -\sum_{\sigma} \int \int \mathrm{d}^3\mathbf{r} \, \mathrm{d}^3\mathbf{R} \, \rho_{\text{el}}(\mathbf{r})\rho_{\text{ion}}(\mathbf{R}) \frac{1}{|\mathbf{r} - \mathbf{R}|} \\ &= -\frac{N}{\Omega} \sum_{\sigma} \int \int \mathrm{d}^3\mathbf{r} \, \mathrm{d}^3\mathbf{R} \, \rho_{\text{el}}(\mathbf{r}) \frac{1}{|\mathbf{r} - \mathbf{R}|} \\ &= -\frac{N}{\Omega} \left[\sum_{\sigma} \int \mathrm{d}^3\mathbf{r} \, \rho_{\text{el}}(\mathbf{r}) \right] \int \mathrm{d}^3\boldsymbol{\xi} \, \frac{1}{|\boldsymbol{\xi}|} = -\frac{N}{\Omega} \, N \lim_{q \to 0} V_q, \end{aligned}$$

where, in the end, we used the fact that the total charge of the electrons is equal to the total charge of the ions, so that $\sum_\sigma \int \mathrm{d}^3\mathbf{r} \, \rho_{\text{el}}(\mathbf{r}) = N$, which is a consequence of the charge neutrality of the system. This means that the electron-ion interaction also diverges in the thermodynamic limit, but it has the opposite sign of the ion-ion interaction; but they do not cancel each other.

Electron-electron interaction for $q = 0$. Moving onto the next interesting term, encoding the electron-electron interaction, we can consider the contribution for $q = 0$, which is the one that diverges, with the idea to show that this divergence is cancelled by the ion-ion and electron-ion interactions. Thus we can express the electron-electron interaction for $q = 0$ as:

$$V_{\text{el-el}}(q = 0) = \frac{1}{2} \sum_{\sigma, \sigma'} \sum_{\mathbf{k}_1, \mathbf{k}_2} \left[\lim_{q \to 0} V_q \right] \hat{c}^\dagger_{\mathbf{k}_1, \sigma} \hat{c}^\dagger_{\mathbf{k}_2, \sigma'} \hat{c}_{\mathbf{k}_2, \sigma'} \hat{c}_{\mathbf{k}_1, \sigma},$$

where now we are interested to change the order of the ladder operators, since we are dealing with fermions, so that the last term becomes:

$$\begin{aligned}
\hat{c}^\dagger_{\mathbf{k}_1,\sigma}\hat{c}^\dagger_{\mathbf{k}_2,\sigma'}\hat{c}_{\mathbf{k}_2,\sigma'}\hat{c}_{\mathbf{k}_1,\sigma} &= -\hat{c}^\dagger_{\mathbf{k}_1,\sigma}\hat{c}^\dagger_{\mathbf{k}_2,\sigma'}\hat{c}_{\mathbf{k}_1,\sigma}\hat{c}_{\mathbf{k}_2,\sigma'} \\
&= -\hat{c}^\dagger_{\mathbf{k}_1,\sigma}\left(\{\hat{c}^\dagger_{\mathbf{k}_2,\sigma'},\,\hat{c}_{\mathbf{k}_1,\sigma}\} - \hat{c}_{\mathbf{k}_1,\sigma}\hat{c}^\dagger_{\mathbf{k}_2,\sigma'}\right)\hat{c}_{\mathbf{k}_2,\sigma'} \\
&= -\hat{c}^\dagger_{\mathbf{k}_1,\sigma}\left(\delta_{\mathbf{k}_1,\mathbf{k}_2}\delta_{\sigma,\sigma'} - \hat{c}_{\mathbf{k}_1,\sigma}\hat{c}^\dagger_{\mathbf{k}_2,\sigma'}\right)\hat{c}_{\mathbf{k}_2,\sigma'},
\end{aligned}$$

where we used the anticommutation relations of the fermionic ladder operators, which state that $\{\hat{c}^\dagger_{\mathbf{k},\sigma}, \hat{c}_{\mathbf{k}',\sigma'}\} = \delta_{\mathbf{k},\mathbf{k}'}\delta_{\sigma,\sigma'}$ and that all the other anticommutators are zero. Thus we can express the electron-electron interaction for $q = 0$ as:

$$\begin{aligned}
V_{\text{el-el}}(q=0) &= \frac{1}{2\Omega}\sum_{\sigma,\sigma'}\sum_{\mathbf{k}_1,\mathbf{k}_2}\left[\lim_{q\to 0} V_q\right]\left(\hat{n}_{\mathbf{k}_1,\sigma}\hat{n}_{\mathbf{k}_2,\sigma'} - \delta_{\mathbf{k}_1,\mathbf{k}_2}\delta_{\sigma,\sigma'}\hat{n}_{\mathbf{k}_1,\sigma}\right) \\
&= \frac{1}{2\Omega}\left[\lim_{q\to 0} V_q\right]\sum_{\sigma}\sum_{\mathbf{k}_1}\left[\left(\sum_{\sigma'}\sum_{\mathbf{k}_2}\hat{n}_{\mathbf{k}_1,\sigma}\hat{n}_{\mathbf{k}_2,\sigma'}\right) - \hat{n}_{\mathbf{k}_1,\sigma}\right],
\end{aligned}$$

and remembering that the sum of the occupation numbers is equal to the total number of particles $\sum_{\mathbf{k}} \hat{n}_{\mathbf{k}} = N$ (independently from $\mathbf{k}_i$, since the sum is over all momenta), we can simplify the expression further

$$V_{\text{el-el}}(q=0) = \frac{1}{2\Omega}(N^2 - N)\lim_{q\to 0} V_q \sim \frac{N^2}{2\Omega}\lim_{q\to 0} V_q,$$

after considering the TD limit, which shows how the electron-electron interaction diverges as $q \to 0$ (a consequence of the long-range nature of the Coulomb interaction).

Cancellation of the divergences. If we sum the ion-ion and ion-electron interactions, we get the following expression:

$$V_{\text{ion-ion}} + V_{\text{ion-el}} = \frac{N^2}{2\Omega}\lim_{q\to 0} V_q - \frac{N^2}{\Omega}\lim_{q\to 0} V_q = -\frac{N^2}{2\Omega}\lim_{q\to 0} V_q,$$

which is easily recognized as the opposite of the electron-electron interaction for $q = 0$ (at the TD limit), so that we have:

$$V_{\text{ion-ion}} + V_{\text{ion-el}} + V_{\text{el-el}}(q=0) = 0,$$

effectively cancelling the divergences of the theory, so that we can get rid of the $q = 0$ contribution to the electron-electron interaction and all the terms related to the interactions with the jellium background; now we can focus on the remaining contributions, which are well defined and do not diverge.

After the removal of the divergences, we can express the **Hamiltonian of the homogeneous electron gas** as:

$$H_{\text{el}} = \sum_{\mathbf{k}\sigma}\frac{\hbar^2 k^2}{2m}\hat{c}^\dagger_{\mathbf{k},\sigma}\hat{c}_{\mathbf{k},\sigma} + \frac{1}{2\Omega}\sum_{\sigma,\sigma'}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{q}\neq 0}\left(\frac{4\pi e^2}{q^2}\right)\hat{c}^\dagger_{\mathbf{k}_1+\mathbf{q},\sigma}\hat{c}^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma'}\hat{c}_{\mathbf{k}_2,\sigma'}\hat{c}_{\mathbf{k}_1,\sigma},$$

where we have neglected the constant term that diverges as $q \to 0$, since it does not affect the properties of the system, along with the terms related to the interactions with the jellium background. This Hamiltonian is the starting point for many-body perturbation theory and other techniques used to study interacting systems, and it is a very important model in condensed matter physics, since it captures many of the essential features of real materials, such as metals and semiconductors.

The strength of this model is that it is simple enough to be solved exactly in some cases, but it is also rich enough to capture many of the essential features of real materials, such as the Fermi surface, the density of states, and the response functions. It is also a very useful model to study the effects of interactions on the properties of the system, such as the formation of collective excitations, the screening of the Coulomb interaction, and the emergence of new phases of matter.

2.2 | Non Interacting Jellium Hamiltonian

Thus the hamiltonian of the electron gas, after having removed all the divergences, can be expressed as:

$$H_{\mathrm{el}} = \sum_{\mathbf{k}\sigma} \frac{\hbar^2 k^2}{2m} \hat{c}^{\dagger}_{\mathbf{k},\sigma} \hat{c}_{\mathbf{k},\sigma} + \frac{1}{2\Omega} \sum_{\sigma,\sigma'} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{q}\neq 0} \left(\frac{4\pi e^2}{q^2}\right) \hat{c}^{\dagger}_{\mathbf{k}_1+\mathbf{q},\sigma} \hat{c}^{\dagger}_{\mathbf{k}_2-\mathbf{q},\sigma'} \hat{c}_{\mathbf{k}_2,\sigma'} \hat{c}_{\mathbf{k}_1,\sigma},$$

where the idea is to use the eigenvalues and eigenvectors of the non-interacting part of the Hamiltonian, which is the kinetic energy term, as a basis on which we will apply perturbation theory to study the effects of the interactions.

The non-interacting part of the Hamiltonian is diagonal in the momentum space, and its eigenvalues are given by:

$$H^{(0)}_{\mathrm{jel}} \Psi_{\mathbf{k},\sigma} = \frac{\hbar^2 k^2}{2m} \Psi_{\mathbf{k},\sigma},$$

(?) check if chi-und-sigma exist

where the eigenvectors are normalized plane waves of the form:

$$\Psi_{\mathbf{k},\sigma}(\mathbf{r}) = \frac{1}{\sqrt{\Omega}} e^{i\mathbf{k}\cdot\mathbf{r}} \chi_{\sigma}.$$

The idea is to use periodic boundary conditions to quantize the momentum, in particular we use the **Born-von Karman boundary conditions**, which state that the wavefunction must be periodic in the volume of the system, so that we have:

$$k_x = \frac{2\pi}{L_x} n_x, \quad k_y = \frac{2\pi}{L_y} n_y, \quad k_z = \frac{2\pi}{L_z} n_z,$$

so that, with the idea to pass to the continuum limit, we can express the sum over the momentum as an integral, so that we have to consider $L_i \to \infty$, such that

$$\sum_{\mathbf{k}} \to \frac{\Omega}{(2\pi)^3} \int \mathrm{d}^3\mathbf{k},$$

where the measure is dimensionless, since the factor Ω has dimensions of volume and the factor $(2\pi)^3$ has dimensions of volume in momentum space, so that the ratio is dimensionless. This is a very important step, since it allows us to pass from a discrete sum to a continuous integral, which is much easier to handle in the thermodynamic limit.

Now we can order the states according to their momenta and spin:

$$|\mathbf{k}_1, \uparrow\rangle, |\mathbf{k}_1, \downarrow\rangle, |\mathbf{k}_2, \uparrow\rangle, |\mathbf{k}_2, \downarrow\rangle, \ldots$$

and in this way we can order the eigenvalues of the non-interacting Hamiltonian from the lowest to the highest, so that we have:

$$\epsilon_{\mathbf{k}_1} < \epsilon_{\mathbf{k}_2} < \ldots$$

Thus if we have N electrons, we can fill the lowest N states according to the Pauli principle, so that we have a **Fermi sea** of electrons, which is the ground state of the non-interacting Hamiltonian, and it is given by:

$$|FS\rangle = \hat{c}^{\dagger}_{\mathbf{k}_{N/2},\uparrow} \hat{c}^{\dagger}_{\mathbf{k}_{N/2},\downarrow} \cdots \hat{c}^{\dagger}_{\mathbf{k}_2,\uparrow} \hat{c}^{\dagger}_{\mathbf{k}_2,\downarrow} \hat{c}^{\dagger}_{\mathbf{k}_1,\uparrow} \hat{c}^{\dagger}_{\mathbf{k}_1,\downarrow} |0\rangle,$$

where $|0\rangle$ is the vacuum state, which is the state with no particles, and the operators $\hat{c}^{\dagger}_{\mathbf{k},\sigma}$ are the creation operators that create an electron with momentum $\mathbf{k}$ and spin σ. The Fermi sea is a very important state, since it is the ground state of the non-interacting Hamiltonian, and it is also a very good approximation to the ground state of the interacting Hamiltonian, since the interactions are weak compared to the kinetic energy.

The fermi energy is the energy of the highest occupied state in the Fermi sea, and it is associated to the Fermi momentum, which is the momentum of the highest occupied state, so that we have:

$$E_F = \frac{\hbar^2 k_F^2}{2m}, \quad k_F = \frac{1}{\hbar}\sqrt{2mE_F},$$

from which we can extract the Fermi velocity, which is the velocity of the electrons at the Fermi surface, so that we have:

$$v_F = \frac{\hbar k_F}{m}.$$

Now how do we connect the the microscopic parameters of the model, such as the density of electrons, to the Fermi energy and the Fermi momentum? We can express the density of electrons as:

$$n = \frac{N}{\Omega} = \frac{1}{\Omega}\langle FS|\hat{N}|FS\rangle = \frac{1}{\Omega}\sum_{\mathbf{k},\sigma}\langle FS|\hat{n}_{\mathbf{k},\sigma}|FS\rangle,$$

where now we can express the sum over the momentum as an integral, so that we have:[2]

$$n = \frac{1}{\Omega}\frac{\Omega}{(2\pi)^3}\sum_\sigma\int \mathrm{d}^3\mathbf{k}\,\langle FS|\hat{n}_{\mathbf{k},\sigma}|FS\rangle = \frac{1}{(2\pi)^3}\sum_\sigma\int \mathrm{d}^3\mathbf{k}\,\theta(k_F-k) = \frac{2}{(2\pi)^3}\int_0^{k_F}k^2\mathrm{d}k\int_{-1}^1\mathrm{d}\cos\theta\int_0^{2\pi}\mathrm{d}\phi = \frac{1}{3\pi^2}k_F^3,$$

where the factor of 2 comes from the spin degeneracy, since we have two spin states for each momentum state, and the integral is over the volume of the Fermi sphere, which is given by:

$$\int_0^{k_F}k^2\mathrm{d}k\int_{-1}^1\mathrm{d}\cos\theta\int_0^{2\pi}\mathrm{d}\phi = \frac{4\pi}{3}k_F^3,$$

so that we can express the Fermi momentum as a function of the density as:

$$k_F = (3\pi^2 n)^{1/3},$$

which is a very important relation, since it allows us to connect the microscopic parameters of the model, such as the density of electrons, to the Fermi energy and the Fermi momentum, which are the parameters that determine the properties of the system.

Example (Some reasonable values for the parameters of the model). Let us consider a typical lattice, for example a cubic lattice with lattice constant $a = 2\,\text{Å}$, which is a typical value for the lattice constant of a metal. We are considering one electron per unit cell, which contains one atom in the center of the unit cell. Thus we are modelling one electron in a volume $\Omega = 8a^3 \sim 8(10\times 10^{-10}\,\mathrm{m}^3) \sim 10^{-29}\,\mathrm{m}^3$.

We can consider copper, which has only one electron in the outer shell, as the other transition metals have more than one electron in the outer shell, so that we can model it as a free electron gas:

$$\mathrm{Cu}: \quad 3d^{10}4s^1, \quad n_{Cu} = 8.47\times 10^{28}\,\mathrm{m}^{-3}.$$

Thus we can compute the Fermi momentum and energy as: (?) check temperature

$$k_F = 13.6\,\mathrm{nm}^{-1}, \quad E_F \sim 7\,\mathrm{eV} \sim 81\,600\,\mathrm{K}.$$

[2] The discretization goes to the continuum and we loose the PBC as we defined them, making just the right integration.......

TODO: put graph of energy density

The Fermi surface is very resilient to temperature, since the Fermi energy is much larger than the thermal energy at room temperature, which is of the order of $\sim 25\,\mathrm{meV}$, so that we can safely assume that the Fermi surface is well defined at room temperature, and that the properties of the system are determined by the electrons at the Fermi surface.

If we now compute the typical velocity of the electrons at the Fermi surface, we get:

$$v_F = \frac{\hbar k_F}{m} \sim 1.5 \times 10^6\,\mathrm{m\,s^{-1}},$$

which is a very high velocity, but only a fraction of the speed of light, so that we can safely assume that the electrons are non-relativistic, and that the non-relativistic quantum mechanics is a good approximation to describe their behavior.

TODO: put image of copper and sodium fermi surfaces

In the surface there are holes: it is a periodic lattice, so that the Fermi surface is not a perfect sphere, but it has some distortions due to the periodic potential of the lattice, which can lead to the formation of holes in the Fermi surface, which are regions where the density of states is very low, and which can have important consequences for the properties of the system, such as the transport properties and the response functions.the surface is connected to the Brillouin zone, which is the primitive cell of the reciprocal lattice, and it can have different shapes depending on the symmetry of the lattice, so that in some cases it can have a very complex shape, which can lead to interesting phenomena such as nesting and van Hove singularities.

Let us come back to the non interacting Hamiltonian of the electron gas, which is given by:

$$H^{(0)}_{jel} = \sum_{\mathbf{k}\sigma} \frac{\hbar^2 k^2}{2m} \hat{c}^\dagger_{\mathbf{k},\sigma} \hat{c}_{\mathbf{k},\sigma},$$

then the spctrum of this Hamiltonian is given by:

$$E^{(0)} = \langle FS| \hat{H}^{(0)}_{jel} |FS\rangle = \sum_{\mathbf{k}\sigma} \frac{\hbar^2 k^2}{2m} \langle FS| \hat{c}^\dagger_{\mathbf{k},\sigma} \hat{c}_{\mathbf{k},\sigma} |FS\rangle,$$

where we can express the sum over the momentum as an integral, and we recognize that the integrand is the occupation number, so that we have:

$$\begin{aligned} E^{(0)} &= \frac{2\hbar^2}{m} \frac{\Omega}{(2\pi)^3} \sum_\sigma \int \mathrm{d}^3\mathbf{k}\, \mathbf{k}^2 \theta(k_F - |\mathbf{k}|) = \dots \\ &= \frac{\Omega}{5\pi^2} \frac{\hbar^2 k_F^5}{2m} = \dots = \frac{3}{5} \Omega \frac{k_F^3}{3\pi^2} E_F, \end{aligned}$$

so that in the end $E^{(0)} = \frac{3}{5} N E_F$, which is the energy of the non-interacting Fermi gas, and it is a very important result, since it gives us a reference point to compare with the energy of the interacting system, and to understand the effects of the interactions on the properties of the system. It makes sense that the energy is proportional to the number of particles and to $\frac{3}{5}E_F$, since the energy of the highest occupied state is E_F, and the average energy of the occupied states is $\frac{3}{5}E_F$, which is a consequence of the fact that the density of states is proportional to the square root of the energy.

In the condensed matter community, this problem was central in the sixties and seventies, since it was the first example of a system of interacting fermions that could be solved exactly, and it was used as a benchmark for many-body perturbation theory and other techniques used to study interacting systems. The idea was now to define a universal and dimensionless parameter that characterizes the density of the system, and to use it to understand the properties of the system as a function of the density. This parameter is called the **Wigner-Seitz radius**, and it is defined as:

$$r_s, \quad n = \frac{1}{\frac{4}{3}\pi(r_s a_0)^3},$$

where the Bohr length a_0 is the length scale that characterizes the size of the electron wavefunction, r_s length instead is the length scale that characterizes the average distance between the electrons, so that it represents 1 electron per sphere of radius a_0 if $r_s = 1$. Otherwise for $r_s \to 0$ it represents a very high density, since we have many electrons in a small volume, while for $r_s \to \infty$ it represents a very low density, since we have few electrons in a large volume. Thus the Wigner-Seitz radius is a very important parameter, since it allows us to characterize the properties of the system as a function of the density, and to understand the effects of the interactions on the properties of the system. It is an inverse measure for density, and it is a very useful parameter since we can write

$$\frac{E^{(0)}}{N} = \frac{3}{5}E_F = \frac{3}{5}\frac{\hbar^2 k_F^2}{2m} = \frac{3}{5}\frac{1}{2}\frac{\hbar^2}{me^2}e^2\left(\frac{a_0^2}{a_0^2}k_F^2\right) = \frac{3}{5}\frac{a_0}{2}\frac{e^2}{a_0^2}k_F^2,$$

where if we compute

$$\frac{1}{n} = \frac{4}{3}\pi(r_s a_0)^3 = \frac{3\pi^2}{k_F} \to r_s = \left(\frac{9\pi}{4}\right)^{1/3}\frac{1}{k_F a_0},$$

so that we can express the energy per particle as a function of the Wigner-Seitz radius as:

$$\frac{E^{(0)}}{N} = \frac{3}{5}\left(\frac{9\pi}{4}\right)^{\frac{2}{3}} \underbrace{\frac{e^2}{2a_0}}_{\text{Rydberg}} \frac{1}{r_s^2} = \frac{2.211\ \text{Ry}}{r_s^2},$$

which shows that the energy per particle of the non-interacting Fermi gas scales as $1/r_s^2$, which is a very important result, since it gives us a reference point to compare with the energy of the interacting system, and to understand the effects of the interactions on the properties of the system as a function of the density.

TODO: graph of energy per particle as a function r-und-s.

Remeber that the Rydberg energy is the energy scale that characterizes the strength of the Coulomb interaction, and it is given by: $1\ \text{Ry} = 13.6\,\text{eV}$.

2.3 | I Order Perturbation Theory

The first important term from perturbation theory is the exchange energy term, which is basically due to the undistinguishability of the electrons, and it represents the energy cost of having indistinguishable particles, since the electrons must obey the Pauli principle, which means that they cannot occupy the same state, so that they must have different momenta or different spins, which leads to a reduction of the energy of the system, since the electrons can avoid each other more effectively.

It is interactions that make the system more stable, even the ones just in the .....

2.3.1 | Electron interactions and I order perturbation theory

$$\frac{E^{(1)}}{N} = \frac{\langle FS| \hat{V}_{\text{el-el}} |FS\rangle}{N} = \frac{1}{2\Omega N} \sum_{\sigma_1,\sigma_2} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{q}\neq 0} \left(\frac{4\pi e^2}{q^2}\right) \langle FS| \hat{c}^\dagger_{\mathbf{k}_1+\mathbf{q},\sigma} \hat{c}^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma'} \hat{c}_{\mathbf{k}_2,\sigma'} \hat{c}_{\mathbf{k}_1,\sigma} |FS\rangle \,,$$

where the only possibility for this term to be non-vanishing is that the operators $\hat{c}^\dagger_{\mathbf{k}_1+\mathbf{q},\sigma}$ and $\hat{c}^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma'}$ create particles in the Fermi sphere, while the operators $\hat{c}_{\mathbf{k}_2,\sigma'}$ and $\hat{c}_{\mathbf{k}_1,\sigma}$ destroy particles in the Fermi sea, so that we have:

- $\mathbf{q} = 0$, which means that the two particles have the same momentum, so that we would create and destroy the same particles, which is not possible since we are dealing with fermions, and indeed this term is neglected directly by the summation over $\mathbf{q} \neq 0$.
- $\mathbf{q} = \mathbf{k}_2 - \mathbf{k}_1$ with $\sigma_1 = \sigma_2$, which means that we create a particle with momentum $\mathbf{k}_2$ and we destroy a particle with momentum $\mathbf{k}_1$, so that we have a non-vanishing contribution to the energy, which is called the exchange energy, since it is due to the exchange of particles between the two states.

TODO: add feynman diagram of the direct interaction and of the exchange interaction with right momentum and spin labels.

So the two particles scatter and after the coulomb interaction they exchange their momenta, but they keep the same spin, since the interaction is spin-independent, so that we have basically an exchange of particles between the two states, which is a consequence of the indistinguishability of the electrons, and it leads to a reduction of the energy of the system, since the electrons can avoid each other more effectively.

These two terms are called the direct and exchange terms, or the Hartree and Fock terms, and they are the first two terms in the perturbation theory expansion of the energy of the system.

TODO: oister diagram for exchange and direct interaction + comments on oister

Now we have to compute the exchange energy, which can be expressed as an integral:

$$\langle FS| \hat{c}^\dagger_{\mathbf{k}_1+\mathbf{q},\sigma} \hat{c}^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma'} \hat{c}_{\mathbf{k}_2,\sigma'} \hat{c}_{\mathbf{k}_1,\sigma} |FS\rangle = \delta_{\sigma_1,\sigma_2} \delta_{\mathbf{k}_1+\mathbf{q},\mathbf{k}_2} \langle FS| \hat{c}^\dagger_{\mathbf{k}_1+\mathbf{q},\sigma_1} \hat{c}^\dagger_{\mathbf{k}_1,\sigma_1} \hat{c}_{\mathbf{k}_1+\mathbf{q},\sigma_1} \hat{c}_{\mathbf{k}_1,\sigma_1} |FS\rangle$$

where the delta functions come from the fact that we have to create and destroy the same particles, so that we have to have the same momentum and the same spin, and the expectation value can be computed using the anticommutation relations of the fermionic operators, so that we have:

$$= -\delta_{\sigma_1,\sigma_2} \delta_{\mathbf{k}_1+\mathbf{q},\mathbf{k}_2} \langle FS| \hat{n}_{\mathbf{k}_1+\mathbf{q},\sigma_1} \hat{n}_{\mathbf{k}_1,\sigma_1} |FS\rangle = -\delta_{\sigma_1,\sigma_2} \delta_{\mathbf{k}_1+\mathbf{q},\mathbf{k}_2} \theta(k_F - |\mathbf{k}_1 + \mathbf{q}|) \theta(k_F - |\mathbf{k}_1|),$$

where the minus sign comes from the anticommutation of the fermionic operators, and the theta functions come from the fact that we have to create and destroy particles in the Fermi sea, so that we have to have momenta that are less than the Fermi momentum. Thus the exchange energy can be expressed as:

$$\frac{E^{(1)}}{N} = -\frac{1}{2\Omega N}\sum_{\sigma}\sum_{\mathbf{k},\mathbf{q}\neq 0}\left(\frac{4\pi e^2}{q^2}\right)\theta(k_F - |\mathbf{k}+\mathbf{q}|)\theta(k_F - |\mathbf{k}|),$$

where we have renamed the momentum $\mathbf{k}_1$ to $\mathbf{k}$ for simplicity. Now we can express the sum over the momentum as an integral

$$\sum_{\mathbf{k},\mathbf{q}\neq 0} \to \frac{\Omega^2}{(2\pi)^6}\int \mathrm{d}^3\mathbf{k}\int \mathrm{d}^3\mathbf{q},$$

but having two theta functions in the integrand makes it a bit more complicated, since we have to consider the intersection of two spheres in momentum space, which is the region where both theta functions are equal to 1, so that we have to compute the volume of the intersection of two spheres of radius k_F centered at $\mathbf{k}$ and $\mathbf{k}+\mathbf{q}$.

TODO: add graph of the intersection of the two spheres in momentum space.

We have to notice that if $\mathbf{q} > 2k_F$ then the intersection of the two spheres is empty, so that the exchange energy vanishes in this case. In the overlapping case, the volume of the intersection of the two spheres can be computed using the formula for the volume of a spherical cap, which is given by: having fixed $\mathbf{q}$ (which can vary in all directions but only for a limited range of module) we can express the volume of the intersection as

$$\mathbf{q} = (q\,|\,\theta_q\,|\,\phi_q), \quad \mathbf{k} = (k\,|\,\theta_k\,|\,\phi_k),$$

$$\theta_K^{min} = 0, \text{ to } k_F\cos(\theta_k^{max}) = \frac{q}{2}, \quad \Longrightarrow \frac{q}{2k_F} < \cos(\theta_k^{max}) < 1,$$

since the two spheres intersect only if the distance between their centers is less than the sum of their radii, which means that we have to have $q < 2k_F$, and for the other piece of the sphere we have:

$$k^{max} = k_F, \quad k^{min}\cos(\theta_k) = \frac{q}{2}, \quad \Longrightarrow \frac{q}{2\cos(\theta_k)} < k < k_F,$$

so that the volume of the intersection of the two spheres can be expressed as:

$$\frac{E^{(1)}}{N} = -\frac{2}{2\Omega N}\frac{\Omega}{(2\pi)^3}\int_{-1}^{1}\mathrm{d}\cos(\theta_q)\int_0^{22\pi}\mathrm{d}\phi_q\int_0^{2k_F}\mathrm{d}q\, q^2\times$$
$$\times\frac{\Omega}{(2\pi)^3}\int_{\frac{q}{2k_F}}^{1}\mathrm{d}\cos(\theta_k)\int_0^{2\pi}\mathrm{d}\phi_k\int_{\frac{q}{2\cos(\theta_k)}}^{k_F}\mathrm{d}k\, k^2\times 2\left(\frac{4\pi e^2}{q^2}\right),$$

where the first factor of 2 comes from the spin degeneracy, since we have two spin states for each momentum state, while the second factor of 2 comes from the fact that we have two equivalent sphere caps, since the intersection of the two spheres is symmetric with respect to the plane that bisects the line that connects their centers. Furthermore, ϕ is not constraint on the integral since it is like a rotation around the axis that connects the centers of the two spheres, so that we can integrate it out.

2.4 | II Order Perturbation Theory

Appendices

A | Notation and Conventions

B | Computations and further developments